

PART D1 PROVISION FOR ESCAPE

Deemed-to-Satisfy Provisions

D1.0 Deemed-to-Satisfy Provisions

(a) Where a *Building Solution* is proposed to comply with the *Deemed-to-Satisfy Provisions*, *Performance Requirements* DP1 to DP6, DP8 and DP9 are satisfied by complying with—

- (i) D1.1 to D1.17, D2.1 to D2.24 and D3.1 to D3.12; and
- (ii) in a building containing an *atrium*, Part G3; and
- (iii) in a building in an *alpine area*, Part G4; and
- (iv) for theatres, *stages* and public halls, Part H1; and
- (v) for public transport buildings, Part H2.

(b) Where a *Building Solution* is proposed as an *Alternative Solution* to the *Deemed-to-Satisfy Provisions* of—

- (i) D1.1 to D1.17, D2.1 to D2.24 and D3.1 to D3.12; and
- (ii) in a building containing an *atrium*, Part G3; and
- (iii) in a building in an *alpine area*, Part G4; and
- (iv) for theatres, *stages* and public halls, Part H1; and
- (v) for public transport buildings, Part H2,

the relevant *Performance Requirements* must be determined in accordance with A0.10.

(c) *Performance Requirement* DP7 must be complied with if lifts are to be used to assist occupants to evacuate a building.

There are no *Deemed-to-Satisfy Provisions* for this *Performance Requirement* in respect of using lifts.

D1.1 Application of Part

The *Deemed-to-Satisfy Provisions* of this Part do not apply to the internal parts of a *sole-occupancy unit* in a Class 2 or 3 building or a Class 4 part of a building.

D1.2 Number of exits required

(a) All buildings — Every building must have at least one *exit* from each *storey*.

(b) Class 2 to 8 buildings — In addition to any *horizontal exit*, not less than 2 *exits* must be provided from the following:

- (i) Each *storey* if the building has an *effective height* of more than 25 m.
- (ii) A Class 2 or 3 building subject to C1.5.

(c) Basements — In addition to any *horizontal exit*, not less than 2 *exits* must be provided from any *storey* if egress from that *storey* involves a vertical rise within the building of more than 1.5 m, unless—

- (i) the *floor area* of the *storey* is not more than 50 m²; and
- (ii) the distance of travel from any point on the floor to a single *exit* is not more than 20 m.

(d) Class 9 buildings — In addition to any *horizontal exit*, not less than 2 *exits* must be provided from the following:

- (i) Each *storey* if the building has a *rise in storeys* of more than 6 or an *effective height* of more than 25 m.

- (ii) Any *storey* which includes a *patient care area* in a Class 9a *health-care building*.
- (iii) Any *storey* that contains sleeping areas in a Class 9c building.
- (iv) Each *storey* in a Class 9b building used as an *early childhood centre*.
- (v) Each *storey* in a primary or secondary *school* with a *rise in storeys* of 2 or more.
- (vi) Any *storey* or *mezzanine* that accommodates more than 50 persons, calculated under D1.13.

NSW D1.2(d)(vii)

(e) Exits from Class 9c buildings and patient care areas in Class 9a health-care buildings — In a Class 9a *health-care building* and a Class 9c building, at least one *exit* must be provided from every part of a *storey* which has been divided into *fire compartments* in accordance with C2.2 or C2.5.

(f) Exits in open spectator stands — In an *open spectator stand* containing more than one tier of seating, every tier must have not less than 2 stairways or ramps, each forming part of the path of travel to not less than 2 *exits*.

(g) Access to exits — Without passing through another *sole-occupancy unit* every occupant of a *storey* or part of a *storey* must have access to—

- (i) an *exit*; or
- (ii) at least 2 *exits*, if 2 or more *exits* are *required*.

D1.3 When fire-isolated stairways and ramps are required

(a) Class 2 and 3 buildings — Every stairway or ramp serving as a *required exit* must be fire-isolated unless it connects, passes through or passes by not more than—

- (i) 3 consecutive *storeys* in a Class 2 building; or
 - (ii) 2 consecutive *storeys* in a Class 3 building,
- and one extra *storey* of any classification may be included if—
- (iii) it is only for the accommodation of motor vehicles or for other ancillary purposes; or
 - (iv) the building has a sprinkler system complying with Specification E1.5 installed throughout; or
 - (v) the *required exit* does not provide access to or egress for, and is separated from, the extra *storey* by construction having—
- (A) an FRL of –/60/60, if non- *loadbearing*; and
 - (B) an FRL of 90/90/90, if *loadbearing*; and
 - (C) no opening that could permit the passage of fire or smoke.

(b) Class 5, 6, 7, 8 or 9 buildings — Every stairway or ramp serving as a *required exit* must be fire-isolated unless—

- (i) in a Class 9a *health-care building* — it connects, or passes through or passes by not more than 2 consecutive *storeys* in areas other than *patient care areas*; or
 - (ii) it is part of an *open spectator stand*; or
 - (iii) in any other case except in a Class 9c building, it connects, passes through or passes by not more than 2 consecutive *storeys* and one extra *storey* of any classification may be included if—
- (A) the building has a sprinkler system complying with Specification E1.5 installed throughout; or
 - (B) the *required exit* does not provide access to or egress for, and is separated from, the extra *storey* by construction having—
- (aa) an FRL of –/60/60, if non- *loadbearing*; and
 - (bb) an FRL of 90/90/90 for Type A construction or 60/60/60 for Type B construction, if *loadbearing*; and
 - (cc) no opening that could permit the passage of fire or smoke.

D1.4 Exit travel distances

(a) Class 2 and 3 buildings—

- (i) The entrance doorway of any *sole-occupancy unit* must be not more than—

- (A) 6 m from an *exit* or from a point from which travel in different directions to 2 *exits* is available; or
- (B) 20 m from a single *exit* serving the *storey* at the level of egress to a road or *open space*; and
- (ii) no point on the floor of a room which is not in a *sole-occupancy unit* must be more than 20 m or from a point at which travel in different directions to 2 *exits* is available.

(b) Class 4 parts of a building — The entrance doorway to any Class 4 part of a building must be not more than 6 m from an *exit* or a point from which travel in different directions to 2 *exits* is available.

(c) Class 5, 6, 7, 8 or 9 buildings — Subject to (d), (e) and (f)—

- (i) no point on a floor must be more than 20 m from an *exit*, or a point from which travel in different directions to 2 *exits* is available, in which case the maximum distance to one of those *exits* must not exceed 20 m; and
- (ii) in a Class 5 or 6 building, the distance to a single *exit* serving a *storey* at the level of access to *open space* may be increased to 30 m.

Vic D1.4(d)

(d) Class 9a buildings — In a *patient care area* in a Class 9a building—

- (i) no point on the floor must be more than 12 m from a point from which travel in different directions to 2 *exits* is available; and
- (ii) the maximum distance to one of those *exits* must not be more than 30 m from the starting point.

(e) Open spectator stands — The distance of travel to an *exit* in a Class 9b building used as an *open spectator stand* must be not more than 60 m.

(f) Assembly buildings — In a Class 9b building other than a *school* or *early childhood centre*, the distance to one of the *exits* may be 60 m if—

- (i) the path of travel from the room concerned to that *exit* is through another area which is a corridor, hallway, lobby, ramp or other circulation space; and
- (ii) the room is smoke-separated from the circulation space by construction having an FRL of not less than 60/60/60 with every doorway in that construction protected by a tight fitting, *self-closing*, solid door not less than 35 mm thick; and
- (iii) the maximum distance of travel does not exceed 40 m within the room and 20 m from the doorway to the *exit*.

D1.5 Distance between alternative exits

Exits that are *required* as alternative means of egress must be—

(a) distributed as uniformly as practicable within or around the *storey* served and in positions where unobstructed access to at least 2 *exits* is readily available from all points on the floor including lift lobby areas; and

(b) not less than 9 m apart; and

(c) not more than—

- (i) in a Class 2 or 3 building — 45 m apart; or
- (ii) in a Class 9a *health-care building*, if such *required exit* serves a *patient care area* — 45 m apart; or
- (iii) in all other cases — 60 m apart; and

(d) located so that alternative paths of travel do not converge such that they become less than 6 m apart.

D1.6 Dimensions of exits and paths of travel to exits

In a *required exit* or path of travel to an *exit*—

- (a) the unobstructed height throughout must be not less than 2 m, except the unobstructed height of any doorway may be reduced to not less than 1980 mm; and
 - (b) the unobstructed width of each *exit* or path of travel to an *exit*, except for doorways, must be not less than—
 - (i) 1 m; or
 - (ii) 1.8 m in a passageway, corridor or ramp normally used for the transportation of patients in be *treatment area* or *ward area*; and
 - (iii) in a *public corridor* in a Class 9c building, notwithstanding (c) and (d)—
 - (A) 1.5 m; and
 - (B) 1.8 m for the full width of the doorway, providing access into a *sole-occupancy unit* or communal bathroom; and
 - (c) if the *storey*, *mezzanine* or *open spectator stand* accommodates more than 100 persons but not more than 200 persons, the aggregate unobstructed width, except for doorways, must be not less than—
 - (i) 1 m plus 250 mm for each 25 persons (or part) in excess of 100; or
 - (ii) 1.8 m in a passageway, corridor or ramp normally used for the transportation of patients in be *treatment area* or *ward area*; and
 - (d) if the *storey*, *mezzanine* or *open spectator stand* accommodates more than 200 persons, the aggregate unobstructed width, except for doorways, must be increased to—
 - (i) 2 m plus 500 mm for every 60 persons (or part) in excess of 200 persons if egress involves a floor level by a stairway or ramp with a gradient steeper than 1 in 12; or
 - (ii) in any other case, 2 m plus 500 mm for every 75 persons (or part) in excess of 200; and
 - (e) in an *open spectator stand* which accommodates more than 2000 persons, the aggregate unobstructed width, except for doorways, must be increased to 17 m plus a width (in metres) equal to the number in excess of 2000 divided by 600; and
 - (f) the unobstructed width of a doorway must be not less than—
 - (i) in *patient care areas* through which patients would normally be transported in beds, if the door provides access to, or from, a corridor of width—
 - (A) less than 2.2 m — 1200 mm; or
 - (B) 2.2 m or greater — 1070 mm,
 and where the doorway is fitted with two leaves and one leaf is secured in the closed position in accordance with D2.21(b)(v), the other leaf must permit an unobstructed opening not less than 800 mm wide; or
 - (ii) in *patient care areas* in a *horizontal exit* — 1250 mm; or
 - (iii) the unobstructed width of each *exit* provided to comply with (b), (c), (d) or (e), minus 250 mm;
- Vic D1.6(f)(iv)
- (iv) in a Class 9c building—
 - (A) 1070 mm where it opens from a *public corridor* to a *sole-occupancy unit*; or
 - (B) 870 mm in other *resident use areas*; or
 - (C) 800 mm in non- *resident use areas*,
 and where the doorway is fitted with two leaves and one leaf is secured in the closed position in accordance with D2.21(b)(v), the other leaf must permit an unobstructed opening not less than 870 mm wide in *resident use areas* and 800 mm wide in non- *resident use areas*; or
 - (v) in any other case except where it opens to a *sanitary compartment* or bathroom — 750 mm
- NSW D1.6(f)(vi)
- (g) the unobstructed width of a *required exit* must not diminish in the direction of travel to a road or *open space*, except where the width is increased in accordance with (b)(ii) or (f)(i); and
 - (h) the *required* width of a stairway or ramp must—

- (i) be measured clear of all obstructions such as handrails, projecting parts of barriers and the like
- (ii) extend without interruption, except for ceiling cornices, to a height not less than 2 m vertically along the nosings of the treads or the floor surface of the ramp or landing.

NSW D1.6(i)

D1.7 Travel via fire-isolated exits

(a) A doorway from a room must not open directly into a stairway, passageway or ramp that is *required* to be fire-isolated unless it is from—

- (i) a *public corridor*, public lobby or the like; or
- (ii) a *sole-occupancy unit* occupying all of a *storey*; or
- (iii) a *sanitary compartment*, airlock or the like.

(b) Each *fire-isolated stairway* or *fire-isolated ramp* must provide independent egress from each *storey* served and discharge directly, or by way of its own *fire-isolated passageway*—

- (i) to a road or *open space*; or
- (ii) to a point—
 - (A) in a *storey* or space, within the confines of the building, that is used only for pedestrian movement, car parking or the like and is open for at least $\frac{2}{3}$ of its perimeter; and
 - (B) from which an unimpeded path of travel, not further than 20 m, is available to a road or *open space*; or
- (iii) into a covered area that—
 - (A) adjoins a road or *open space*; and
 - (B) is open for at least $\frac{1}{3}$ of its perimeter; and
 - (C) has an unobstructed clear height throughout, including the perimeter openings, of not less than 3 m; and
 - (D) provides an unimpeded path of travel from the point of discharge to the road or *open space* of not more than 6 m.

(c) Where a path of travel from the point of discharge of a fire-isolated *exit* necessitates passing within 6 m of any part of an *external wall* of the same building, measured horizontally at right angles to the path of travel, that part of the wall must have—

- (i) an FRL of not less than 60/60/60; and
 - (ii) any openings protected internally in accordance with C3.4,
- for a distance of 3 m above or below, as appropriate, the level of the path of travel, or for the height of the wall, whichever is the lesser.

(d) If more than 2 access doorways, not from a *sanitary compartment* or the like, open to a *required* fire-isolated *exit* in the same *storey*—

- (i) a smoke lobby in accordance with D2.6 must be provided; or
- (ii) the *exit* must be pressurised in accordance with AS/NZS 1668.1.

(e) A ramp must be provided at any change in level less than 600 mm in a *fire-isolated passageway* in a Class 9 building.

D1.8 External stairways or ramps in lieu of fire-isolated exits

(a) An external stairway or ramp may serve as a *required exit* in lieu of a fire-isolated *exit* serving a *storey* below an *effective height* of 25 m, if the stairway or ramp is—

- (i) *non-combustible* throughout; and
- (ii) protected in accordance with (c) if it is within 6 m of, and exposed to any part of the *external* building it serves.

(b) For the purposes of this clause—

- (i) exposure under (a)(ii), is measured in accordance with Clause 2.1 of Specification C1.1, as if a building element and the *external wall* of the building was a *fire-source feature* to the *exit*, e: the FRL *required* in Clause 2.1(a)(i) must not be less than 60/60/60; and
- (ii) the plane formed at the construction edge or perimeter of an unenclosed building or part such *deck carpark, open spectator stand* or the like, is deemed to be an *external wall*; and
- (iii) openings in an *external wall* and openings under (c) and (d), are determined in accordance w

(c) The protection referred to in (a)(ii), must adequately protect occupants using the *exit* from exposure to a fire within the building, in accordance with one of the following methods:

- (i) The part of the *external wall* of the building to which the *exit* is exposed must have—
 - (A) an FRL of not less than 60/60/60; and
 - (B) no openings less than 3 m from the *exit* (except a doorway serving the *exit* protected by a –/60/30 fire door in accordance with C3.8(a)); and
 - (C) any opening 3 m or more but less than 6 m from the *exit*, protected in accordance with C3.4 and if wall wetting sprinklers are used, they are located internally.
- (ii) The *exit* must be protected from—
 - (A) any part of the *external wall* of the building having an FRL of less than 60/60/60; and
 - (B) any openings in the *external wall*,
 by the construction of a wall, roof, floor or other shielding element as appropriate in accordance with (d).

(d) The wall, roof, floor or other shielding element *required* by (c)(ii) must—

- (i) have an FRL of not less than 60/60/60; and
- (ii) have no openings less than 3 m from the *external wall* of the building (except a doorway served protected by a –/60/30 fire door in accordance with C3.8(a)); and
- (iii) have any opening 3 m or more but less than 6 m from any part of the *external wall* of the building protected in accordance with C3.4 and if wall wetting sprinklers are used, they are located on exposed to the *external wall*.

D1.9 Travel by non-fire-isolated stairways or ramps

(a) A non- *fire-isolated stairway* or non- *fire-isolated ramp* serving as a *required exit* must provide a continuous means of travel by its own *flights* and landings from every *storey* served to the level at which egress to a road or *open space* is provided.

(b) In a Class 2, 3 or 4 building, the distance between the doorway of a room or *sole-occupancy unit* and the point of egress to a road or *open space* by way of a stairway or ramp that is not fire-isolated and is *required* to serve that room or *sole-occupancy unit* must not exceed—

- (i) 30 m in a building of Type C construction; or
- (ii) 60 m in all other cases.

(c) In a Class 5, 6, 7, 8 or 9 building, the distance from any point on a floor to a point of egress to a road or *open space* by way of a *required* non- *fire-isolated stairway* or non- *fire-isolated ramp* must not exceed 80 m.

(d) In a Class 2, 3 or 9a building, a *required* non- *fire-isolated stairway* or non- *fire-isolated ramp* must discharge at a point not more than—

- (i) 15 m from a doorway providing egress to a road or *open space* or from a *fire-isolated passageway* leading to a road or *open space*; or
- (ii) 30 m from one of 2 such doorways or passageways if travel to each of them from the non- *fire stairway* or non- *fire-isolated ramp* is in opposite or approximately opposite directions.

(e) In a Class 5 to 8 or 9b building, a *required* non- *fire-isolated stairway* or non- *fire-isolated ramp* must discharge at a point not more than—

- (i) 20 m from a doorway providing egress to a road or *open space* or from a *fire-isolated passageway* leading to a road or *open space*; or
- (ii) 40 m from one of 2 such doorways or passageways if travel to each of them from the non- *fire stairway* or non- *fire-isolated ramp* is in opposite or approximately opposite directions.

(f) In a Class 2 or 3 building, if 2 or more *exits* are *required* and are provided by means of internal non- *fire-isolated stairways* or non- *fire-isolated ramps*, each *exit* must—

- (i) provide separate egress to a road or *open space*; and
- (ii) be suitably smoke-separated from each other at the level of discharge.

D1.10 Discharge from exits

(a) An *exit* must not be blocked at the point of discharge and where necessary, suitable barriers must be provided to prevent vehicles from blocking the *exit*, or access to it.

(b) If a *required exit* leads to an *open space*, the path of travel to the road must have an unobstructed width throughout of not less than—

- (i) the minimum width of the *required exit*; or
- (ii) 1 m,

whichever is the greater.

(c) If an *exit* discharges to *open space* that is at a different level than the public road to which it is connected, the path of travel to the road must be by—

- (i) a ramp or other incline having a gradient not steeper than 1:8 at any part, or not steeper than *required* by the *Deemed-to-Satisfy Provisions* of Part D3; or
- (ii) except if the *exit* is from a Class 9a building, a stairway complying with the *Deemed-to-Satisfy* of the BCA.

(d) The discharge point of alternative *exits* must be located as far apart as practical.

(e) In a Class 9b building which is an *open spectator stand* that accommodates more than 500 persons, a *required stairway* or *required ramp* must not discharge to the ground in front of the stand.

NSW D1.10(f)

(f) In a Class 9b building containing an auditorium which accommodates more than 500 persons, not more than $\frac{2}{3}$ of the *required* width of *exits* must be located in the main entrance foyer.

D1.11 Horizontal exits

(a) *Horizontal exits* must not be counted as *required exits*—

- (i) between *sole-occupancy units*; or
- (ii) in a Class 9b building used as an *early childhood centre*, primary or secondary *school*.

(b) In a Class 9a *health-care building* or Class 9c building, *horizontal exits* may be counted as *required exits* if the path of travel from a *fire compartment* leads by one or more *horizontal exits* directly into another *fire compartment* which has at least one *required exit* which is not a *horizontal exit*.

(c) In cases other than in (b), *horizontal exits* must not comprise more than half of the *required exits* from any part of a *storey* divided by a *fire wall*.

(d) *Horizontal exits* must have a clear area on the side of the *fire wall* to which occupants are evacuating, to accommodate the total number of persons (calculated under D1.13) served by the *horizontal exit* of not less than—

- (i) 2.5 m² per patient/resident in a Class 9a *health-care building* or Class 9c building; and
- (ii) 0.5 m² per person in any other case.

(e) Where a *fire compartment* is provided with only two *exits*, and one of those *exits* is a *horizontal exit*, the clear area *required* by (d) is to be of a size that accommodates all the occupants from the *fire compartment* being evacuated.

(f) The clear area *required* by (d) must be connected to the *horizontal exit* by an unobstructed path that has at least the dimensions *required* for the *horizontal exit* and may include the area of the unobstructed path.

D1.12 Non-required stairways, ramps or escalators

An escalator, moving walkway or non- *required* non *fire-isolated stairway* or pedestrian ramp—

(a) must not be used between *storeys* in—

- (i) a *patient care area* in a Class 9a *health-care building*; or
- (ii) a *resident use area* in a Class 9c building; and

(b) may connect any number of *storeys* if it is—

- (i) in an *open spectator stand* or indoor sports stadium; or
- (ii) in a *carpark* or an *atrium*; or
- (iii) outside a building; or
- (iv) in a Class 5 or 6 building that is sprinklered throughout, where the escalator, walkway, stairway complies with Specification D1.12; and

(c) except where permitted in (b) must not connect more than—

- (i) 3 *storeys* if each of those *storeys* is provided with a sprinkler system complying with Specification D1.12 throughout; or
- (ii) 2 *storeys*,

provided that in each case, those *storeys* must be consecutive, and one of those *storeys* is situated at a level at which there is direct egress to a road or *open space*; and

(d) except where permitted in (b) or (c), must not connect, directly or indirectly, more than 2 *storeys* at any level in a Class 5, 6, 7, 8 or 9 building and those *storeys* must be consecutive.

D1.13 Number of persons accommodated

The number of persons accommodated in a *storey*, room or *mezzanine* must be determined with consideration to the purpose for which it is used and the layout of the *floor area* by—

(a) calculating the sum of the numbers obtained by dividing the *floor area* of each part of the *storey* by the number of square metres per person listed in Table D1.13 according to the use of that part, excluding spaces set aside for—

- (i) lifts, stairways, ramps and escalators, corridors, hallways, lobbies and the like; and
- (ii) service ducts and the like, *sanitary compartments* or other ancillary uses; or

(b) reference to the seating capacity in an *assembly building* or room; or

(c) any other suitable means of assessing its capacity.

NSW Table D1.13

Table D1.13 AREA PER PERSON ACCORDING TO USE

Type of use		m ² per person
Art gallery, exhibition area, museum		4
Bar	—bar standing	0.5
	—other	1
Board room		2
Boarding house		15
Cafe, church, dining room		1
<i>Carpark</i>		30
Computer room		25
Court	—judicial area	10
	room —public seating	1
Dance floor		0.5
Dormitory		5
<i>Early childhood centre</i>		4
Factory—	(a) machine shop, fitting shop or like place for cutting, for cutting, grading, finishing or fitting of metals or glass, except in the fabrication of structural steelwork or manufacture of vehicles or bulky products	5
	(b) areas used for fabrication and processing other than those in (a)	50
	(c) a space in which the layout and natural use of fixed plant or equipment determines the number of persons who will occupy the space during working hours	Area per person determined by the use of the plant or equipment
Gymnasium		3
Hostel, hotel, motel, guest house		15
Indoor sports stadium—arena		10
Kiosk		1
Kitchen, laboratory, laundry		10
Library	—reading space	2
	—storage space	30
Office, including one for typewriting or document copying		10
<i>Patient care areas</i>		10
Plant room—ventilation, electrical or other service units		30
	—boilers or power plant	50
Reading room		2
Restaurant		1
<i>School</i>	—general classroom	2
	—multi-purpose hall	1
	—staff room	10
	—trade and —primary	4
	practical —secondary	As for workshop
Shop	—space for sale of goods—	
	(a) at a level entered direct from the open air or any lower level	3
	(b) all other levels	5
Showroom—display area, covered mall or arcade		5
Skating rink, based on rink area		1.5
Spectator stand, audience viewing area:		
	—standing viewing area	0.3
	—removable seating	1
	—fixed seating (number of seats)	
	—bench seating (450 mm/person)	
Storage space		30

Type of use	m ² per person
Swimming pool, based on pool area	1.5
Switch room, transformer room	30
Telephone exchange	30
—private	
Theatre and public hall	1
Theatre dressing room	4
Transport terminal	2
Workshop —for maintenance staff	30
—for manufacturing processes	As for Factory
Note: Bar standing is the area used by standing patrons and extends not less than 1.5m wide from the outside edge of the bar top for the length of the serving area of the bar.	

D1.14 Measurement of distances

The nearest part of an *exit* means in the case of—

- (a) a *fire-isolated stairway*, *fire-isolated passageway*, or *fire-isolated ramp*, the nearest part of the doorway providing access to them; and
- (b) a non- *fire-isolated stairway*, the nearest part of the nearest riser; and
- (c) a non- *fire-isolated ramp*, the nearest part of the junction of the floor of the ramp and the floor of the *storey*; and
- (d) a doorway opening to a road or *open space*, the nearest part of the doorway; and
- (e) a *horizontal exit*, the nearest part of the doorway.

D1.15 Method of measurement

The following rules apply:

- (a) In the case of a room that is not a *sole-occupancy unit* in a Class 2 or 3 building or Class 4 part of a building, the distance includes the straight-line measurement from any point on the floor of the room to the nearest part of a doorway leading from it, together with the distance from that part of the doorway to the single *required exit* or point from which travel in different directions to 2 *required exits* is available.
- (b) Subject to (d), the distance from the doorway of a *sole-occupancy unit* in a Class 2 or 3 building or a Class 4 part of a building is measured in a straight line to the nearest part of the *required single exit* or point from which travel in different directions to 2 *required exits* is available.
- (c) Subject to (d), the distance between *exits* is measured in a straight line between the nearest parts of those *exits*.
- (d) Only the shortest distance is taken along a corridor, hallway, external balcony or other path of travel that curves or changes direction.
- (e) If more than one corridor, hallway, or other internal path of travel connects *required exits*, the measurement is along the path of travel through the point at which travel in different directions to those *exits* is available.
- (f) If a wall (including a demountable *internal wall*) that does not bound—
 - (i) a room; or

(ii) a corridor, hallway or the like,
causes a change of direction in proceeding to a *required exit*, the distance is measured along the path of travel past that wall.

(g) If permanent fixed seating is provided, the distance is measured along the path of travel between the rows of seats.

(h) In the case of a non- *fire-isolated stairway* or non- *fire-isolated ramp*, the distance is measured along a line connecting the nosings of the treads, or along the slope of the ramp, together with the distance connecting those lines across any intermediate landings.

D1.16 Plant rooms, lift machine rooms and electricity network substations: Concession

(a) A ladder may be used in lieu of a stairway to provide egress from—

- (i) a plant room with a *floor area* of not more than 100 m²; or
- (ii) all but one point of egress from a plant room, a lift machine room or a Class 8 *electricity network substation* with a *floor area* of not more than 200 m².

(b) A ladder permitted under (a)—

- (i) may form part of an *exit* provided that in the case of a *fire-isolated stairway* it is contained with or
- (ii) may discharge within a *storey* in which case it must be considered as forming part of the path and
- (iii) for a plant room or a Class 8 *electricity network substation*, must comply with AS 1657; and
- (iv) for a lift machine room, where access is provided from within a machine room to a secondary rung type ladder complying with AS 1657 may be used, provided that—
 - (A) the height between the floors is not more than 2800 mm; and
 - (B) the ladder is inclined at an angle to the horizontal not less than 65 degrees nor more than 75 degrees; and
 - (C) the distance between the front face of the ladder and any adjacent obstruction is not less than—
 - (aa) 960 mm, where the ladder is inclined 65 degrees to the horizontal; or
 - (bb) 760 mm, where the ladder is inclined 75 degrees to the horizontal; or
 - (cc) a distance that is determined by interpolating the values in (aa) and (bb), where the ladder is inclined at any angle between 65 degrees and 75 degrees to the horizontal; and
 - (D) a clear space not less than 600 mm exists between the foot of the ladder and any equipment.

D1.17 Access to lift pits

Access to lift pits must—

(a) where the pit depth is not more than 3 m, be through the lowest landing doors; or

(b) where the pit depth is more than 3 m, be provided through an access doorway complying with the following:

- (i) In lieu of D1.6, the doorway must be level with the pit floor and not be less than 600 mm wide 1980 mm high clear opening, which may be reduced to 1500 mm where it is necessary to connect to a compressed buffer.
- (ii) No part of the lift car or platform must encroach on the pit doorway entrance when the car is at the compressed buffer.
- (iii) Access to the doorway must be by a stairway complying with AS 1657.
- (iv) In lieu of D2.21, doors fitted to the doorway must be—
 - (A) of the horizontal sliding or outwards opening hinged type; and

(B)self-closing and self-locking from the outside; and

(C)marked on the landing side with the letters not less than 35 mm high:

“DANGER LIFTWELL â” ENTRY OF UNAUTHORIZED PERSONS PROHIBITED â” KEEP CLEAR AT ALL TIMES”

ACT D1.101, ACT D1.102